Experiment 1

Case Study on the Analysis of SDLC Models Applied to a Real-World Software Application

Application Chosen: Online Food Delivery System

Aim

The aim of this experiment is to study and analyze the Software Development Life Cycle (SDLC) models applied to an **Online Food Delivery System**. This case study helps understand how different SDLC models can be used in developing a real-world software application. It also compares various SDLC models based on their development approach, advantages, and limitations. The study identifies the most suitable SDLC model for developing an efficient, secure, and user-friendly food delivery application.

Objectives

The objectives of this case study are to understand the importance of SDLC in software development and to analyze different SDLC models. It aims to compare the Waterfall, Incremental, Spiral, Agile, and DevOps models based on their features and applications. The study also identifies the best SDLC model for an Online Food Delivery System. Another objective is to understand how proper planning and development improve software quality, reduce risks, and increase customer satisfaction.

Introduction

An **Online Food Delivery System** is a software application that allows customers to browse restaurants, order food, make online payments, and track deliveries in real time. Restaurants can manage menus, receive orders, and update order status, while delivery partners can view assigned deliveries and update delivery progress. Such applications require regular updates, secure payment processing, and high system reliability.

Therefore, selecting the right SDLC model is essential for successful software development and maintenance.

About the Online Food Delivery System

The Online Food Delivery System connects customers, restaurants, and delivery partners through a single platform. Customers can search for restaurants, view menus, place orders, make digital payments, and track their orders. Restaurant owners can update menu items, accept or reject orders, and monitor sales. Delivery personnel receive delivery requests and update the delivery status. The application provides convenience, reduces waiting time, and improves customer satisfaction.

Functional Requirements

The application should provide all the necessary features for customers, restaurants, delivery partners, and administrators. Customers should be able to register, log in, search for restaurants, place food orders, make online payments, and track deliveries. Restaurants should manage menus and orders, while delivery partners should update delivery status. Administrators should monitor users, restaurants, payments, and reports to ensure smooth operation

Module	**Functions**
Customer	Registration, Login, Food Ordering
Restaurant	Menu Management, Order Processing
Delivery	Delivery Tracking, Status Update
Payment	Online Payment, Transaction History
Admin	User & Restaurant Management

SDLC Models Used in Software Development

Different SDLC models are available to develop software based on project requirements. The Waterfall model follows a sequential approach, while the Incremental model develops the application in smaller modules. The Spiral model focuses on risk management during development. Agile supports continuous development through short iterations and customer feedback. DevOps combines software development and deployment using automation, making software delivery faster and more reliable.

Model	Main Feature
Waterfall	Sequential Development
Incremental	Module-wise Development
Spiral	Risk Analysis
Agile	Iterative Development
DevOps	Continuous Integration & Deployment

Analysis of SDLC Models

The Waterfall model is suitable for projects with fixed requirements but does not easily accommodate changes. The Incremental model delivers the software in smaller parts, allowing users to access basic features earlier. The Spiral model is useful for complex projects because it focuses on identifying and reducing risks. Agile encourages continuous customer feedback and frequent software updates, making it highly flexible. DevOps improves collaboration between development and operations teams, enabling faster releases and better system maintenance.

Why Agile is Suitable for the Online Food Delivery System

The Agile model is the most suitable choice for an Online Food Delivery System because customer needs and market trends change frequently. New features such as discount offers, payment options, restaurant listings, and delivery tracking can be added during each development sprint. Agile allows developers to release updates quickly while continuously collecting customer feedback. This approach improves software quality,

reduces development risks, and ensures the application remains competitive and user-friendly.

SDLC Phases Applied to the Application

The development of the Online Food Delivery System follows all the SDLC phases. During the planning phase, project objectives and feasibility are defined. Requirement analysis identifies customer and restaurant needs. The design phase creates system architecture and user interfaces. Development involves coding the application modules, followed by testing to identify and fix defects. Finally, the software is deployed for users, and regular maintenance ensures smooth performance and feature updates.

Comparison of SDLC Models

SDLC Model	Advantages	Disadvantages
Waterfall	Easy to manage and document	Difficult to change requirements
Incremental	Faster delivery of features	Requires good planning
Spiral	Excellent risk management	Expensive and time-consuming
Agile	Flexible and customer-focused	Needs active customer involvement
DevOps	Faster deployment and automation	Requires skilled professionals

Results

The analysis shows that different SDLC models are suitable for different types of software projects. For an Online Food Delivery System, the Agile model provides the greatest flexibility because it supports continuous development and customer feedback. Features can be updated frequently without affecting the entire application. Agile also improves software quality, reduces development time, and helps deliver better user

experiences. Therefore, Agile is considered the most suitable SDLC model for this application.

Conclusion

This case study explains how various SDLC models can be applied to the development of an Online Food Delivery System. Each SDLC model has its own strengths and limitations depending on project requirements. After comparing different models, Agile is identified as the most appropriate approach because it supports iterative development, frequent testing, and continuous improvements. Selecting the right SDLC model helps deliver high-quality software that satisfies customer needs and adapts to changing market demands.